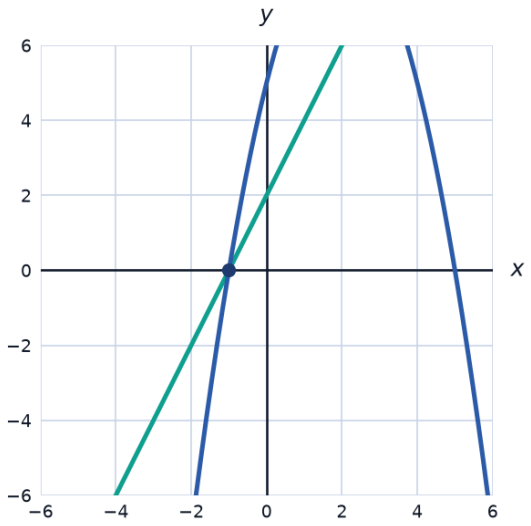
Indices, Surds & Quadratics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The points A, B, and C have coordinates A(2, 5), B(k, 1), and C(8, -3). If A, B, and C are collinear, find the value of k.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of $y = f(x)$ is a curve passing through the points A(1, 2), B(4, 2), and C(2, 5). This graph is then subjected to a sequence of transformations. First, it is reflected in the line $y = x$. Let the image of the points A, B, and C under this reflection be A', B', and C' respectively. Secondly, the graph of the reflected curve is translated by the vector $(-3; 0)$. Let the final image of the points A', B', and C' after this translation be A'', B'', and C''. Given that the original curve is a parabola with its axis of symmetry parallel to the y-axis, and passes through A, B, and C, find the coordinates of C''.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows the graph of the function $y = f(x)$. The coordinates of the points A, B, and C are given. The graph is transformed by a translation of 3 units to the right and 2 units down. What are the new coordinates of point A?

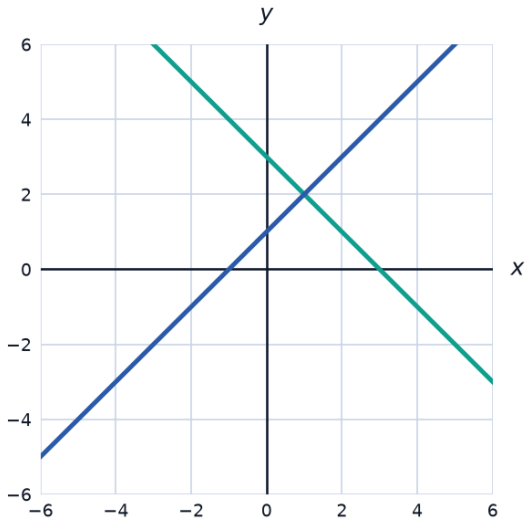
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5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given the quadratic equation $x^2 - 5x + 6 = 0$, find the sum of the roots.

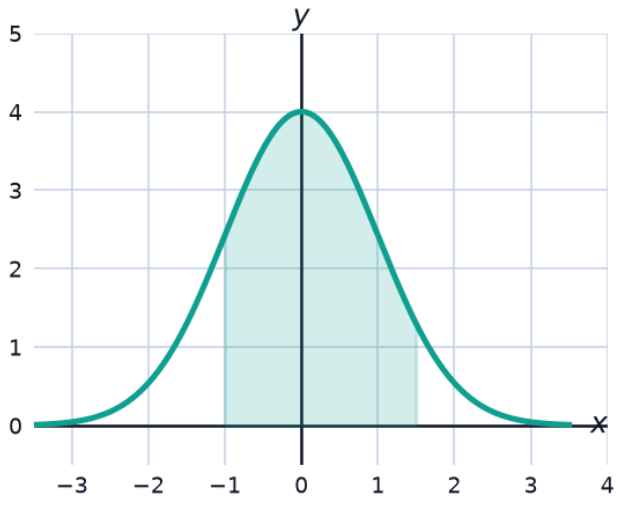
6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The weights of bags of sugar are normally distributed with a mean of 1000 g and a standard deviation of 5 g. Let X be the weight of a randomly selected bag of sugar. Calculate the probability that a bag of sugar weighs between 990 g and 1010 g.

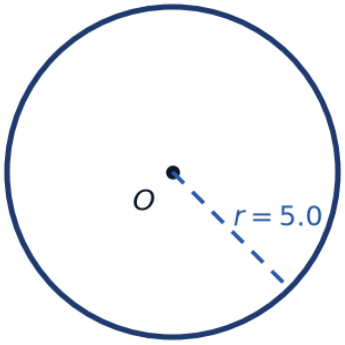


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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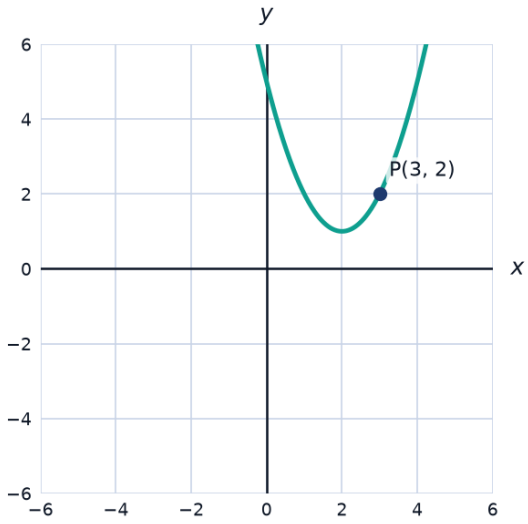
8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows a sector of a circle with centre O. The arc length of the sector is given by 6π cm and the area of the sector is 27π cm^2. The radius of the circle is r cm and the angle of the sector is θ radians. Find the value of r and θ.



9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

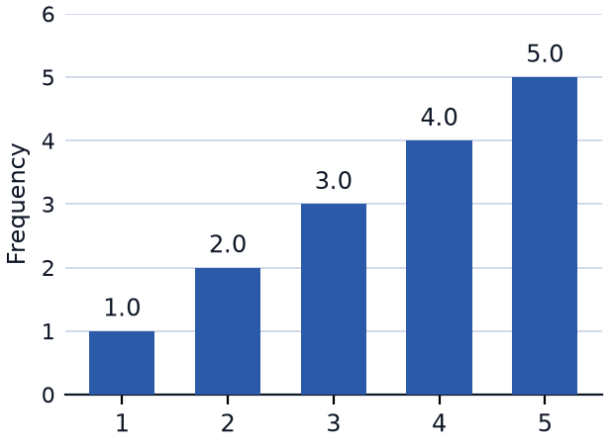
The diagram shows part of the curve with equation $y = x^2 - 4x + 5$ and the point $P(3, 2)$ lying on the curve. Find the gradient of the tangent to the curve at the point P.



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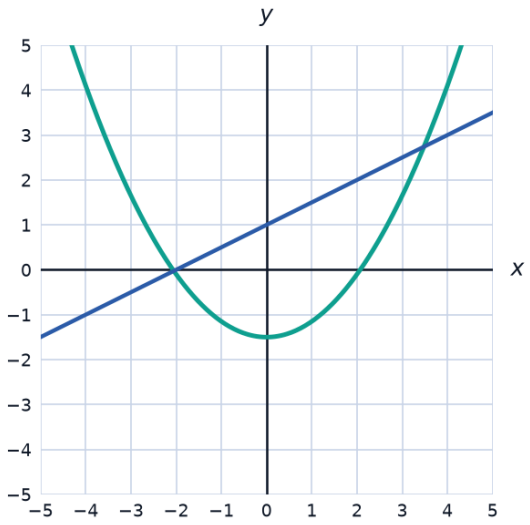
10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A discrete random variable, X, can take the values 1, 2, 3, 4, and 5. The probability distribution of X is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4, 5\}$, where k is a constant. Find the value of k.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

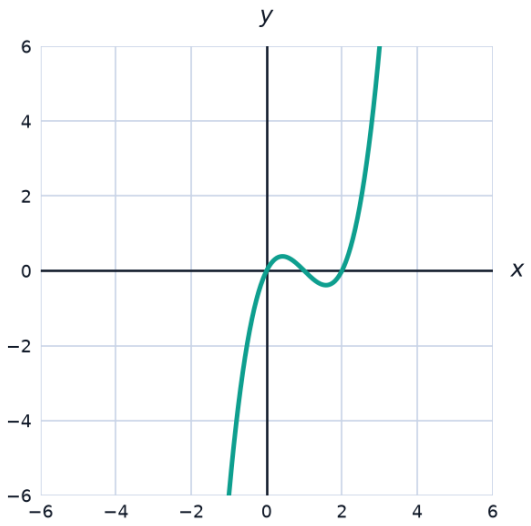
The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD 6$cmandtheradiusisUSD3$ cm, find the angle θ in radians.



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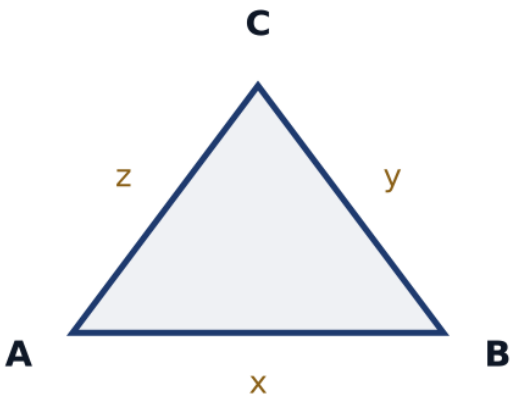
12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A curve is given by the equation $y = x^3 - 3x^2 + 2x$. Find the gradient of the curve at $x = 2$.



13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows a triangle ABC with coordinates A(2, 1), B(8, 1) and C(5, 5). Find the length of the side AC.



14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A fair coin is tossed 3 times. Let X be the number of heads obtained. What is the probability that X = 2?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Differentiation & Integration

A particle's velocity v m/s at time t seconds is given by $v = \frac{1}{2}t^2 - 3t + 5$. The displacement of the particle from a fixed origin is s metres. It is known that at $t = 2$, the displacement $s = 7$. The displacement function $s(t)$ can be expressed in the form $s(t) = At^3 + Bt^2 + Ct + D$. Find the value of A and the value of D.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

The first three terms of the binomial expansion of $(1 + kx)^n$ are $1 + 6x + 12x^2$. Find the value of k.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Functions: Composite & Inverse

A particle of mass 8 kg is moving under the action of a net force $\mathbf{F}$ N. The acceleration of the particle, $\mathbf{a}$ m/s^2, is given by the vector $\mathbf{a} = (2t - 4;\, t^2 - 9)$ for $t \geq 0$, where t is the time in seconds. The particle is initially at rest at the origin.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Circular Measure

A sector of a circle with radius r cm has a perimeter of P cm. The area of the sector is A cm^2. Prove that $P^2 \geq 16A$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Series & Binomial Theorem

A particle of mass 5 kg moves under the action of a variable force $\mathbf{F}$ N. The force is given by $\mathbf{F} = (3x^2 - 2x)\mathbf{i}$ N, where x is the displacement in metres from a fixed origin O. The particle moves from $x = 2$ m to $x = 4$ m. Find the work done by the force $\mathbf{F}$ during this displacement.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Kinematics

Solve the quadratic equation $2x^2 - 5x - 3 = 0$ by completing the square. State the vertex of the corresponding parabola $y = 2x^2 - 5x - 3$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify $\left(\frac{8x^6}{27y^3}\right)^{1/3}$ completely.

a. $\frac{2x^2}{3y}$

b. $\frac{4x^2}{9y}$

c. $\frac{2x^3}{3y}$

d. $\frac{8x^2}{27y}$

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify $\sqrt{48} - \sqrt{12} + \sqrt{75}$ into the form $k\sqrt{3}$.

a. $7\sqrt{3}$

b. $5\sqrt{3}$

c. $9\sqrt{3}$

d. $6\sqrt{3}$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Rationalize the denominator of $\frac{6}{3-\sqrt{3}}$ and write in simplest form.

a. $3 + \sqrt{3}$

b. $2 + \sqrt{3}$

c. $3 - \sqrt{3}$

d. $6 + 2\sqrt{3}$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $2x^2 - 12x + 11$ in the form $a(x-h)^2 + k$.

a. $2(x-3)^2 - 7$

b. $2(x-3)^2 + 11$

c. $2(x-6)^2 - 25$

d. $2(x-3)^2 - 11$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the minimum value of $f(x) = 3x^2 + 18x + 20$ and the value of x at which it occurs.

a. Minimum is -7 at $x = -3$

b. Minimum is 20 at $x = 0$

c. Minimum is -7 at $x = 3$

d. Minimum is 7 at $x = -3$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The curve $y = -(x+4)^2 + 15$ has a stationary point. State its nature and coordinates.

a. Maximum at $(-4, 15)$

b. Minimum at $(-4, 15)$

c. Maximum at $(4, 15)$

d. Minimum at $(4, -15)$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the set of values of k for which the equation $x^2 + kx + (k+3) = 0$ has two distinct real roots.

a. $k < -2$ or $k > 6$

b. $-2 < k < 6$

c. $k \leq -2$ or $k \geq 6$

d. $k > 6$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The line $y = 2x + c$ is tangent to the curve $y = x^2 - 4x + 7$. Find the exact value of c.

a. -2

b. 2

c. 7

d. -6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Determine the condition on p such that the quadratic $px^2 + 4x + p = 0$ has no real roots, where $p > 0$.

a. $p > 2$

b. $0 < p < 2$

c. $p > 4$

d. $p < -2$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the disguised quadratic equation $x^4 - 5x^2 - 36 = 0$ for all real solutions.

a. $x = \pm 3$

b. $x = \pm 2$

c. $x = \pm 3, \pm 2i$

d. $x = 9$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A rectangular enclosure is formed using 80 meters of fencing against a straight stone wall. What is the maximum possible area of the enclosure in square meters?

a. $800\ \text{m}^2$

b. $400\ \text{m}^2$

c. $1600\ \text{m}^2$

d. $600\ \text{m}^2$

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the discriminant of $3x^2 - 5x - 10 = 0$ and state the number of real roots.

a. $\Delta = -145$: two distinct real roots

b. $\Delta = 145$: one repeated real root

c. $\Delta = 153$: no real roots

d. $\Delta = 145$: two distinct real roots

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows a right-angled triangle ABC, with the right angle at D. The altitude from D to the hypotenuse BC is drawn. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

a. 6 cm

b. 5 cm

c. 7 cm

d. 8 cm

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows a right-angled triangle ABC, with the right angle at C. The altitude from C to the hypotenuse AB is CD. Given that $AD = 4$ and $DB = 9$, find the length of the altitude CD.

a. 3

b. 4

c. 5

d. 6

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Rationalise the denominator of $\frac{1}{\sqrt{23}}$.

a. $\frac{\sqrt{23}}{23}$

b. $\frac{1\sqrt{23}}{1}$

c. $\frac{1\sqrt{23}}{23}$

d. $1\sqrt{23}$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Write $a^{-3} \times a^{-1}$ in the form a^k, giving the value of k.

a. k=-4

b. k=-3

c. k=4

d. k=-2

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Simplify $\sqrt{96}$, giving your answer in the form $a\sqrt{b}$ with b as small as possible.

a. $4\sqrt{24}$

b. $5\sqrt{6}$

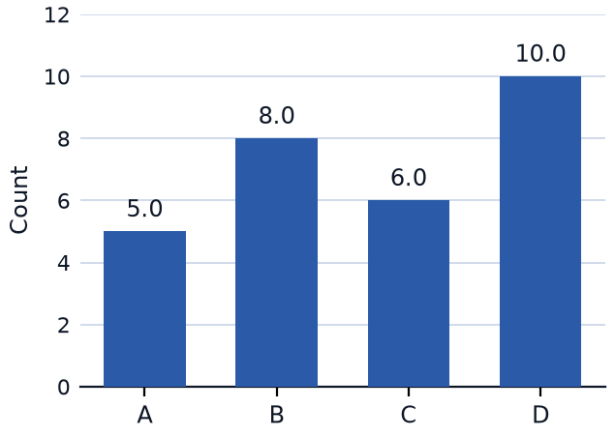
c. $4\sqrt{6}$

d. $4\sqrt{7}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Circular Measure

The histogram below shows the distribution of heights of a group of students. The total number of students in the group is 50. If the frequency density for the class interval $60 \le h < 70$ is 1.2, find the value of k.



a. 10

b. 12

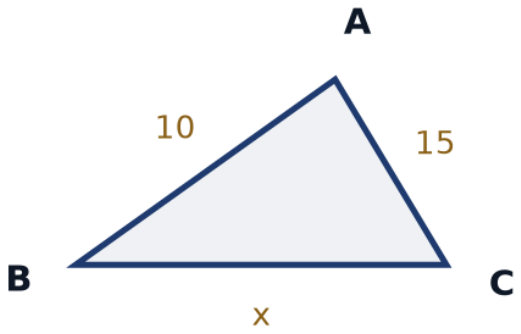
c. 15

d. 20

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Series & Binomial Theorem

In triangle ABC, angle ABC = θ degrees, and angle ACB = 2θ degrees. The lengths of sides AB and AC are 10 cm and 15 cm respectively. The length of BC is denoted by x cm. Given that the area of triangle ABC is 60 cm^2, find the value of x, to 3 significant figures.



a. 13.7

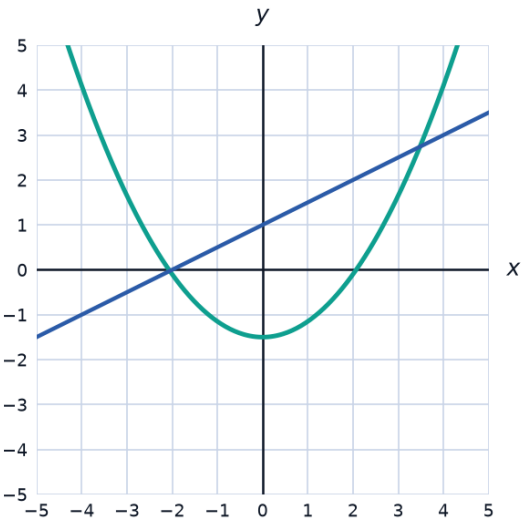
b. 11.4

c. 12.8

d. 14.9

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Kinematics

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is 6 cm and the radius is 3 cm, find the angle θ in radians.



a. $\frac{1}{2}$ radians

b. 2 radians

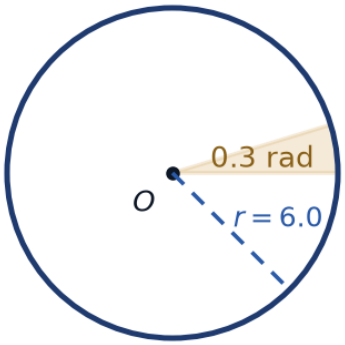
c. $\frac{2}{3}$ radians

d. $\frac{3}{2}$ radians

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 12.01 Forces & Equilibrium

A sector has radius 6 cm and angle 0.3 rad. Find the arc length.



a. 9/5

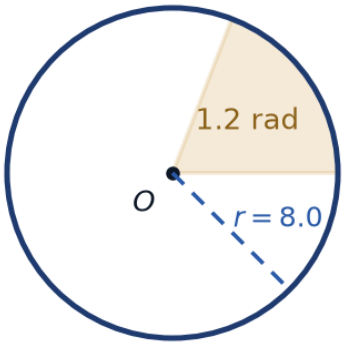
b. 63/10

c. 14/5

d. 27/5

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 13.01 Newton's Laws

A sector OAB has radius 8 cm and angle $\angle AOB = 1.2$ rad. Find the area of the minor segment cut off by chord AB, leaving your answer in exact form.



a. $32(1.2 - \sin(1.2))$

b. $64(1.2 - \sin(1.2))$

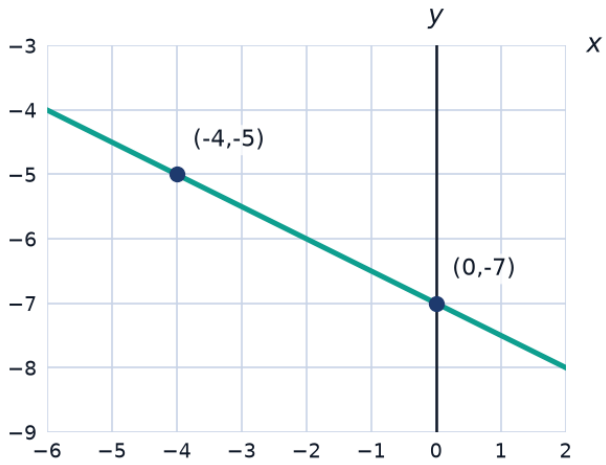
c. $33(1.2 - \sin(1.2))$

d. $32(1.2 + \sin(1.2))$

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 14.01 Energy, Work & Power

Find the equation of the line through $(0, -7)$ and $(-4, -5)$, in the form $y = mx + c$.



a. y=-1/2x+7

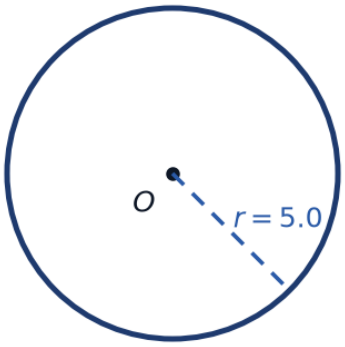
b. y=-1/2x-7

c. y=-1/2x-6

d. y=1/2x-7

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 15.01 Representation of Data

A circle has equation $(x - 7)^2 + (y - 5)^2 = 25$. State its centre and radius.



a. centre $(5, 7)$, radius 5

b. centre $(7, 5)$, radius 5

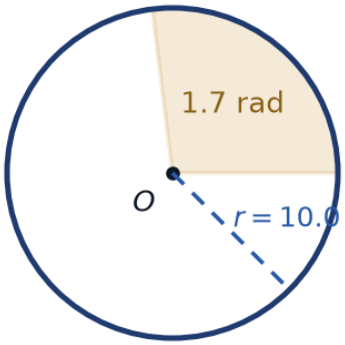
c. centre $(-7, -5)$, radius 5

d. centre $(7, 5)$, radius 25

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 16.01 Permutations & Combinations

A sector has radius 10 cm and angle 1.7 rad. Find its area.



a. 170

b. 95

c. 85

d. 17

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